Artificial Intelligence
Advanced Topics in AI & ML
Final Exam

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ML Research



Content

1. Exam information

Content

1. Exam information
2. Exam problems

Exam information

- Exam format: **hand-written on paper**
- Exam duration: **30 min**
- Materials allowed to be used: **hand-written notes (1 sheet of paper, both sides can be used))**
- No one is allowed to use either the **phone**, **laptop** or **printed materials**
- Exam parts:
  1. Short questions about the course content: 100%

Proctor Policy I

- All students taking in-person exams are expected to follow academic integrity standards to ensure a fair and secure testing environment. Only materials explicitly permitted by the professor may be used during the exam. Any authorized materials, including cheat sheets, must be clearly defined by the professor in advance (for example, one handwritten page) and may be briefly checked at the start of the exam to ensure compliance.
- Students arriving late may be admitted during a limited grace period, typically 10–15 minutes after the exam begins. After that time, entry may be restricted in order to maintain exam integrity and minimize disruption. Students who arrive late will not receive additional testing time.
- All personal belongings, including bags, phones, smartwatches, headphones, and unauthorized notes, must be placed in a designated area before the exam begins. The designated area may include an adjacent seat or underneath the student's seat, as directed by the professor/proctor. All electronic devices must be completely powered off, not simply silenced or placed on vibrate mode.

Proctor Policy II

- Students must follow all instructions provided by the instructor or proctor and remain seated unless permission is granted to leave the room. Communication with other students is prohibited at all times during the exam.
- Short restroom breaks may be permitted one student at a time at the discretion of the professor/proctor. Students may not take any exam materials, notes, or electronic devices with them during a break. Exam time will not be paused or extended for restroom breaks.
- Any behavior suggesting academic misconduct, including but not limited to looking at another student's work, communicating during the exam, accessing unauthorized materials, or attempting to use concealed electronic devices, will be treated as suspected cheating. Suspected violations will be documented and reported in accordance with institutional academic integrity policies and procedures. Consequences for confirmed violations may include disciplinary action as outlined by the institution.

Exam process

Exam procedure:

- Exam Assignment will be open in **Canvas** for the **time of Exam** (and **only** during this time!)
- You'll need to write Exam Assignment on paper sheets and when you finish, take a photo / convert to pdf and **submit** as a 'File Upload' to Canvas
- You'll have only **1 attempt** to submit, so please be careful!
- After submission, please **deliver** the written sheets to **Professor / Classroom Manager** (to be collected, stored, and probably scanned for better quality)

Remarks on process

- You need to add your name at the top left of the each side of every sheet
- You need to add the page number at the top right of the each side of every sheet (1, 2, 3, 4, 5, ...)
- Allowed: Only one person at a time may leave the room

Exam topics: concepts

- AI History overview
- Deep Learning and Neural Nets approach
- GenerativeAI concept (and its connection to DL and NN)
- GenAI: GAN-based and Diffusion-based approaches
- Transformer properties and Transformer-based encoders (BERT) and decoders (GPT)
- LLM concept (and its connection to Transformers)
- DL applications: Computer Vision (and Convolution Neural Nets)
- DL applications: Automatic Speech Recognition (and Recurrent Neural Nets)
- Multi-tasking concept (and its connection to optimization)
- Meta-learning (including few-shot learning) framework
- Multi-modality concept (including architectural considerations)
- Neural Net Robustness problem
- Adversarial attacks (including digital and real-world)
- EmbodiedAI main principle
- Autonomous Driving Automation levels
- Autonomous Driving Stack

Thank you *all*!